

4 (**short**) In the circuit of Fig. 4, the op-amp is ideal except for its finite gain, A . Derive an expression for the gain of the circuit, v_o / v_i and show that as A tends to infinity, the gain of the circuit tends to

$$\frac{v_o}{v_i} = 1 + \frac{R_1}{R_2}$$

[10]

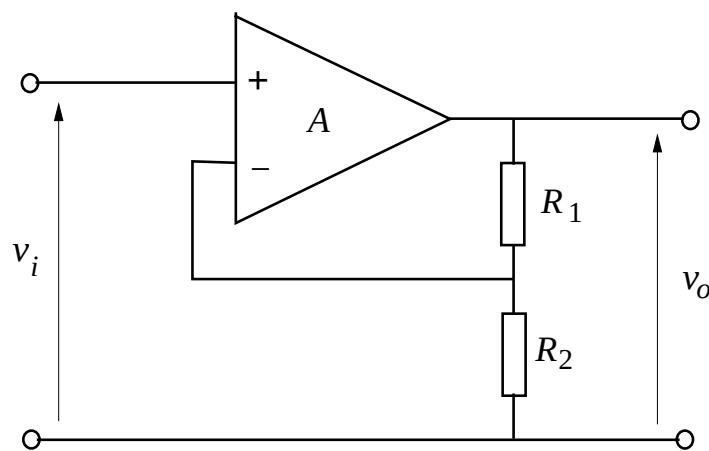


Fig. 4